



4 A curve has parametric equations

$$x = t^3 - t^2 + t - 1 \quad \text{and} \quad y = te^t.$$

(a) Show that 1 is the only real value of t for which $x = 0$. [1]

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(b) Show that $\frac{dy}{dx} = \frac{(t+1)e^t}{3t^2 - 2t + 1}$. [3]

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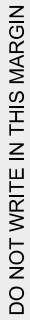
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